# Six boat counts

What holds a paper boat?

Name _____ Date _____

All numbers are pretend. Show your tens and ones or write your steps.

**1.** Little gathers 28 pennies, then 17 more beside the tub. How many pennies are ready?

28 + 17 = _____

Show your thinking.

**2.** Little has 52 pennies beside the tub and puts 26 away. How many remain?

52 − 26 = _____

Show your thinking.

**3.** Little counts 36 folds across several boats, then 27 more folds. How many folds in all?

36 + 27 = _____

Show your thinking.

# Six boat counts

What holds a paper boat?

Name _____ Date _____

All numbers are pretend. Show your tens and ones or write your steps.

**4.** Little draws 71 drops beside a boat sketch, then erases 38. How many drawn drops remain?

71 − 38 = _____

Show your thinking.

**5.** Little draws 24 waterline marks, then follows “Add 19 marks.” How many marks are drawn?

24 + 19 = _____

Show your thinking.

**6.** A pretend boat voyage lasts 63 minutes. After 28 minutes, how many minutes remain?

63 − 28 = _____

Show your thinking.

## A sheet with a secret

Little Wonderer set a sheet of paper beside a tub. "Could this carry a penny?" Big Wonderer put down a short list of folding commands.





## The folding commands

"Fold the paper in half. Fold the corners down." Little followed each command in order. More folds made an open boat with walls and a broad bottom.





## The first voyage

Little lowered the boat into the tub. It floated: it stayed at the water's surface. The boat pushed water aside, or displaced it.





## A climbing line

Little added a penny, then another. This load was the weight the boat carried. The boat sank lower, displacing more water.





## One penny too many

"Add one penny. Watch the waterline," Little wrote. Then one more penny made the rim dip under. Water spilled inside. The boat was swamped!





## A soft ending

Later, the boat sagged. Paper's tiny plant fibers absorb water: they take it in. Wet paper grows heavy and soft until its folds lose their shape.



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





WONDER DAILY

# What holds a paper boat?

# Give the boat an instruction.

What holds a paper boat? · Curiosity day

Name _____ Date _____

Choose one scene. Write one clear command sentence for Little's boat. You may add a second step.



The folding commands



A climbing line



A soft ending

Draw the boat before or after your command. Add one label.

fold · boat · tub · penny · waterline · load · absorb · first · then

First, ... tell how to begin folding

Then, ... tell what to add or watch

Next time, ... tell what to change or compare

Read your sentence. Does it tell what to do, using a command such as Fold, Add, Watch, or Compare?

# What holds a paper boat?

## 01 Can the shape of paper help it carry a penny?



An imagined scene. Penny counts and time in the lessons are pretend.

**Look closer.** Compare the flat sheet with the folded boat in the next scene.

**A little explanation.** Folded walls and a broad, flat bottom make a space that keeps water out. The air inside is part of what floats. A flat sheet may rest on water briefly, but it has no walls to protect that space when a load pushes it down.

**Try this.** Imagine folding the sides up. Which part would keep water out?

## 02 What is holding the boat up?



An imagined scene. Penny counts and time in the lessons are pretend.

**Look closer.** Look where the water meets the boat. The bottom of the tub is far below.

**A little explanation.** The boat displaces water: it pushes water aside. Water pushes up with a force as big as the weight of the water pushed aside. The boat floats when that upward push equals the boat's weight.

**Try this.** Use your voice to follow the steps: push water aside, feel water push up, float.

## 03 Why does the waterline climb?



An imagined scene. Penny counts and time in the lessons are pretend.

**Look closer.** The pennies are inside. The water reaches higher on the sides.

**A little explanation.** Pennies add load, the weight the boat carries. To balance that extra weight, the boat sinks lower and pushes aside more water. Too much load, or a tip, lets water spill over the sides and swamp the boat.

**Try this.** Predict what would happen to the waterline if Little removed a penny.

## 04 Why can't a paper boat float forever?



An imagined scene. Penny counts and time in the lessons are pretend.

**Look closer.** The damp paper has sagged. Its crisp folds are softer.

**A little explanation.** Paper is made of tiny plant fibers that absorb water. Wet paper gets heavier and softer and loses its shape. A crayon or wax coat slows the soaking, but it does not make the boat last forever.

**Try this.** Choose one command for another voyage, such as "Watch the waterline."